# 字符串入门

## 人员

王纯硕 崔孝泽 屈钰轩 陈奕圻 赵庭澍 彭俊文 到课

## 上周作业检查

上周作业链接: https://cppoj.kids123code.com/contest/4030

比赛概况 题目列表 选择题列表 提交记录 实时榜单 选择题排行榜 编辑比赛

**王向东老师周六三点半C++二维数组练习2**

刷新

| # | 用户名 | 姓名 | 编程分 | 时间 | A | B | C | D | E |
|---|---|---|---|---|---|---|---|---|---|
| 1 | quyuxuan | 屈钰轩 | 500 | 45 | 100 | 100 | 100 | 100 | 100 |
| 2 | wangchunshuo | 王纯硕 | 500 | 46 | 100 | 100 | 100 | 100 | 100 |
| 3 | zhaotingshu | 赵庭澍 | 500 | 47 | 100 | 100 | 100 | 100 | 100 |
| 4 | xiejuntian | 解钧添 | 500 | 48 | 100 | 100 | 100 | 100 | 100 |
| 5 | zhangjiarui | 张嘉芮 | 450 | 45 | 100 | 100 | 100 | 100 | 50 |
| 6 | chenyiqi | 陈奕圻 | 400 | 46 | 100 | 100 | 100 | 100 | |
| 7 | cuixiaoze | 崔孝泽 | 400 | 54 | 0 | 100 | 100 | 100 | 100 |
| 8 | pengjunwen | 彭俊文 | 300 | 45 | | 100 | 100 | 100 | |

## 作业

https://cppoj.kids123code.com/contest/4163 (课上讲了 A ~ H 题, 作业是 A ~ H 题必做)

## 课堂表现

今天上课给同学们讲解了 char 和 string 的简单应用, 同学们课上整体吸收还不错, 课下要加强复习。

## 课堂内容

**判断是什么字符**

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
  char c;
  cin >> c;
  if (c>='A' && c<='Z') {
    cout << "upper";
  }
  else if (c>='a' && c<='z') {
    cout << "lower";
```

```
  }
  else {
    cout << "digit";
  }
  return 0;
}
```

### 大小写转换

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
  char c;
  cin >> c;
  if (c>='A' && c<='Z') {
    cout << char(c-'A'+'a');
  }
  else {
    cout << char(c-'a'+'A');
  }
  return 0;
}
```

### 打印小写字母表

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
  for (char i = 'a'; i <= 'm'; i++) {
    cout << i;
  }
  cout << endl;
  for (char i = 'n'; i <= 'z'; i++) {
    cout << i;
  }
  cout << endl;
  for (char i = 'z'; i >= 'n'; i--) {
    cout << i;
  }
  cout << endl;
  for (char i = 'm'; i >= 'a'; i--) {
    cout << i;
```

```
  }
  return 0;
}
```

### 石头剪子布

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
  int n;
  cin >> n;
  for (int i = 1; i <= n; i++) {
    string a, b;
    cin >> a >> b;
    if (a == b) {
      cout << "Tie" << endl;
    }
    else if ((a=="Rock"&&b=="Scissors") || (a=="Scissors"&&b=="Paper") ||
(a=="Paper"&&b=="Rock")) {
      cout << "Player1" << endl;
    }
    else {
      cout << "Player2" << endl;
    }
  }
  return 0;
}
```

### 判断字符串是否为回文

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
  string a, b;
  cin >> a;
  b = a;
  reverse(b.begin(), b.end());
  if (a == b) {
    cout << "yes" << endl;
  }
  else {
    cout << "no" << endl;
```

```
  }
  return 0;
}
```

**【深基6.例1】自动修正**

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
  string s;
  cin >> s;
  int n = s.size();
  s = " " + s;
  for (int i = 1; i <= n; i++) {
    if (s[i]>='a' && s[i]<='z') {
      s[i] = s[i]-'a'+'A';
    }
  }
  for (int i = 1; i <= n; i++) {
    cout << s[i];
  }
  return 0;
}
```

**配对碱基链**

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
  string s;
  cin >> s;
  int n = s.size();
  s = " " + s;
  for (int i = 1; i <= n; i++) {
    if (s[i] == 'A') {
      cout << "T";
    }
    else if (s[i] == 'T') {
      cout << "A";
    }
    else if (s[i] == 'G') {
      cout << "C";
```

```
    }
    else {
      cout << "G";
    }
  }
  return 0;
}
```

### 基因相关性

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
  double t;
  cin >> t;

  string a, b;
  cin >> a >> b;
  int n = a.size();
  a = " " + a, b = " " + b;

  int cnt = 0;
  for (int i = 1; i <= n; i++) {
    if (a[i] == b[i]) {
      cnt++;
    }
  }

  double z = 1.0*cnt/n;
  if (z >= t) {
    cout << "yes" << endl;
  }
  else {
    cout << "no" << endl;
  }
  return 0;
}
```